Module Splitter: An Adaptive Semantic Tree Structuring

A Major Project Report

Submitted by

Nitish	220120101112
Nitin Kumar	220120101111
Mohd. Aasif	220120101101
Himanshi Pandey	730120101012
Sonu Kumar	220120101162

Under the Supervision of

Mr. Upendra Verma
Assistant Professor

Department of Computer Science & Engineering

In Partial Fulfillment of the Requirements

for the Degree of

Bachelor of Technology

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

TULA'S INSTITUTE, DEHRADUN

(Affiliated to VMSB Uttarakhand Technical University, Dehradun)

DECLARATION

We, declare that the work embodied in this Project report is our own original work carried out by us under the supervision of **Mr. Upendra Verma** (Assistant Professor ,Dept. of CSE), for the session 2025-26 at Tula's Institute, Dehradun. The matter embodied in this Project report has not been submitted elsewhere for the award of any other degree. We declare that we have faithfully acknowledged, given credit to and referred to the researchers wherever the work has been cited in the text and the body of the report. We further certify that we have not wilfully lifted up some other's work, para, text, data, results, etc. reported in the journals, books, magazines, reports, dissertations, thesis or available at websites and have included them in this project report and cited as our own work.

Date: 24/05/2026

Place: Dehradun

Nitish

Nitin Kumar

Mohd. Aasif

Himanshu Pandey

Sonu Kumar

Certificate from the Supervisor/Co-supervisor

This is to certify that the Project Report entitled:

"Module Splitter: An Adaptive Semantic Tree Structuring"

Submitted by:

Nitish	220120101112
Nitin Saini	220120101111
Mohd. Aasif	220120101101
Himanshu Pandey	730120101012
Sonu Kumar	220120101162

at Tula's Institute, Dehradun for the degree of **Bachelor of Technology** is their original work carried out by them under my guidance and supervision. This work is fully or partially has not been submitted for the award of any other degree or diploma. The assistance and help taken during the course of the study has been duly acknowledged and the source of literature amply recorded.

Supervisor Signature :

Supervisor Name : Mr. Upendra Verma

Supervisor Designation : Assistant Professor

Date : 24/05/26

ACKNOWLEDGEMENT

The successful completion of this project has been made possible through the support, guidance, and encouragement of several distinguished individuals, to whom we wish to express our deepest gratitude. We begin by acknowledging the grace of the Almighty and the unwavering support of our families, whose blessings and sacrifices have been a constant source of strength throughout our academic journey. We extend our sincere and profound gratitude to **Mr. Upendra Verma, Assistant Professor, Department of CSE, Tula's Institute**, for his visionary leadership, sustained mentorship, and the academic environment he has fostered, which made this project possible.

We are deeply thankful to **Mr. Sandeep Kumar, Head of Department (CSE)**, for his continuous academic support, constructive feedback, and for providing the necessary resources and infrastructure required for the execution of this project. We would like to express our heartfelt appreciation to **Mr. Sharad Pratap Singh, Project Coordinator**, for his diligent coordination, timely guidance, and dedication toward ensuring the smooth progress of our project work.

We owe special thanks to our project mentor, whose expert supervision, technical insight, and patient guidance were instrumental at every stage of this project. His encouragement and commitment to our growth as engineers have been truly inspiring.

Finally, we are grateful to all faculty members, technical staff, and peers who, directly or indirectly, contributed to the successful completion of this project.

Date: 24/05/2026

Submitted by:

Nitish	220120101112
Nitin Saini	220120101111
Mohd. Aasif	220120101101
Himanshu Pandey	730120101012
Sonu Kumar	220120101162

TABLE OF CONTENTS

LIST OF SYMBOLS

S.No	Symbol	Meaning
1	(S_{AB})	Set of shared symbols between entities, files, or regions
2	(k_B)	Kind-specific multiplier applied to score calculations for the target entity (B)
3	(\sigma) (sigma)	Extraction score for a region; represents the decision engine output
4	(\Delta MI)	Predicted Maintainability Index improvement after extraction
5	CC	Cyclomatic Complexity; measures independent execution paths
6	CogCC	Cognitive Complexity; nesting-aware complexity metric
7	MI	Maintainability Index; composite maintainability score
8	HV	Halstead Volume; information-theoretic size metric
9	LOC	Lines of Code used in metric and weighting formulas
10	(n_1)	Number of distinct operators in Halstead metrics
11	(n_2)	Number of distinct operands in Halstead metrics
12	(N_1)	Total occurrences of operators in Halstead metrics
13	(N_2)	Total occurrences of operands in Halstead metrics
14	(V) (Vocabulary)	Halstead vocabulary, calculated as (n_1 + n_2)
15	(L) (Length)	Halstead length, calculated as (N_1 + N_2)
16	Effort	Halstead Effort, calculated as Difficulty × Volume
17	Difficulty	Halstead Difficulty, calculated as ((n_1/2) \times (N_2/n_2))

LIST OF ABBREVIATIONS

ABBREVIATION	MEANING
AST	Abstract Syntax Tree
ASTra	Adaptive Semantic Tree Restructuring
SCC	Strongly Connected Component
CC	Cyclomatic Complexity
CogCC	Cognitive Complexity
MI	Maintainability Index
HV	Halstead Volume
LOC	Lines Of Code
API	Application Programming Interface
DP	Dependency Propagation
LRU	Least Recently Used
CSP	Content Security Policy
Jest	JavaScript Testing Framework
Vitest	Vite-native Testing Framework
TS	TypeScript
TSX	TypeScript + JSX
JSX	JavaScript XML
webview	VS Code Embedded Browser Panel Used for the Analysis UI
WorkspaceEdit	VS Code API Object Representing Atomic Edits

ABSTRACT

Large-scale front-end software systems frequently experience gradual architectural degradation as projects evolve. Over time, individual source files begin accumulating multiple unrelated responsibilities, leading to highly coupled and difficult-to-maintain modules. In modern JavaScript and TypeScript ecosystems, it is increasingly common to find React components, business logic, custom hooks, utility methods, constants, configuration objects, type definitions, asynchronous handlers, and styling logic residing within the same file. While such accumulation initially accelerates development velocity, it eventually produces oversized modules that are difficult to understand, test, refactor, and scale collaboratively. Existing refactoring approaches remain heavily dependent on manual developer intervention and rarely provide objective, measurable criteria for deciding when and how decomposition should occur. As a result, software teams often postpone restructuring efforts until maintainability costs become severe.

Module Splitter — the Adaptive Semantic Tree Restructuring Framework, an automated static-analysis-driven system designed to identify, evaluate, and semantically decompose overloaded front-end modules into maintainable architectural units. Implemented as a Visual Studio Code extension named **astra-extension**, the framework provides end-to-end automated restructuring support for TypeScript, JavaScript, JSX, and TSX codebases. Unlike traditional code formatting or linting tools that focus primarily on style violations, Module Splitter performs deep semantic analysis of source files using Abstract Syntax Tree (AST) traversal, dependency mapping, contextual scoring, and framework-aware heuristics to generate a fully resolved decomposition plan.

The framework introduces a ten-stage adaptive analysis pipeline capable of transforming monolithic modules into modularized structures while preserving behavioral correctness and project consistency. The pipeline begins with lexical preprocessing and AST generation, followed by symbol graph construction, dependency extraction, complexity evaluation, semantic region detection, and extraction candidate scoring. It then proceeds through dynamic threshold calibration, dependency-safe extraction planning, import/export reconciliation, workspace-wide conflict analysis, and final atomic application through the VS Code Workspace Edit API. This design allows the framework to operate directly within the developer workflow without requiring external build tools or repository restructuring utilities.

A primary contribution of this research is the introduction of an exact AST-walk implementation of the **SonarSource Cognitive Complexity specification** adapted specifically for modern front-end architectures. Existing implementations often approximate cognitive complexity using simplified nesting metrics, leading to inaccurate measurements in component-driven applications. Module Splitter instead performs fine-grained traversal across conditional branches, recursive closures, asynchronous control structures, JSX expressions, higher-order callbacks, and nested rendering logic to compute precise cognitive complexity values at both file and region levels. This allows the framework to identify semantically dense regions that significantly reduce code readability and developer comprehension.

The system further introduces **ExtractionOracle**, a multi-factor extraction scoring engine that evaluates potential decomposition regions using eight weighted analytical dimensions. These dimensions include cognitive complexity, dependency isolation, identifier cohesion, lexical similarity, export viability, reusability probability, side-effect density, and framework boundary separation. Each candidate region receives a composite extraction score derived through weighted normalization and adaptive balancing strategies. Rather than relying on simplistic line-count thresholds, the framework determines extraction suitability using a contextual scoring methodology that reflects real-world maintainability concerns.

To improve decision accuracy across heterogeneous repositories, the framework incorporates a **Halstead-calibrated dynamic threshold model**. Traditional refactoring tools typically apply fixed thresholds globally, causing over-extraction in lightweight modules and under-extraction in complex enterprise codebases. Module Splitter instead computes adaptive thresholds using Halstead metrics, operator-operand distributions, vocabulary density, and regional complexity gradients. This enables extraction boundaries to scale dynamically according to the structural characteristics of each file and repository. The result is significantly improved decomposition consistency across varying coding styles and architectural patterns.

Another major contribution is the introduction of an **incremental hash-keyed region cache** for repeated analysis optimization. Large front-end repositories often contain thousands of files, making continuous static analysis computationally expensive during active development. To address this issue, the framework computes structural hashes for semantic regions and stores intermediate analysis artifacts in a persistent cache layer. During subsequent executions, unchanged regions bypass redundant analysis stages, allowing the framework to reuse previously computed dependency graphs, complexity values, and extraction candidates.

Experimental evaluation demonstrates that this mechanism reduces repeated analysis latency by approximately ten times compared to full recomputation pipelines, significantly improving responsiveness within interactive development environments.

The framework additionally constructs a **cross-file workspace dependency graph** capable of resolving direct imports, indirect references, transitive re-export chains, and alias-based module relationships across the entire project. This graph enables Module Splitter to safely determine extraction destinations while preventing circular dependency introduction and namespace conflicts. The system can also identify structurally similar utility modules and recommend merge targets for extracted regions, thereby supporting not only decomposition but also architectural consolidation. Such workspace-level reasoning distinguishes the framework from localized refactoring tools that analyze files in isolation without considering broader repository structure.

Recognizing the diversity of modern front-end ecosystems, the framework incorporates **framework-specific smell detection plugins** for React, Vue 3, Angular, and Svelte applications. Each plugin introduces ecosystem-aware heuristics capable of identifying architecture-specific anti-patterns. For React projects, the system detects oversized component trees, hook overload concentration, render-logic interleaving, and excessive prop drilling regions. Vue analysis includes composable misuse detection, reactive state overloading, and template-script imbalance analysis. Angular plugins identify service inflation, decorator clustering, and module dependency anomalies, while Svelte integrations analyze reactive statement density and component-state entanglement. These plugins allow Module Splitter to adapt extraction strategies according to framework conventions rather than applying uniform language-level heuristics.

The implementation architecture of Module Splitter is designed around modular analysis stages and deterministic transformation generation. Source files are parsed using high-fidelity TypeScript compiler APIs, enabling precise preservation of formatting, comments, decorators, generics, and type annotations during extraction. The framework generates fully resolved split plans that include newly created files, corrected relative import paths, updated export declarations, and automatically generated barrel index files. Optional test scaffolds are produced for extracted modules based on detected framework conventions and project testing configurations. All modifications are applied atomically using the VS Code Workspace Edit API, ensuring consistency and rollback safety in the event of transformation conflicts.

To evaluate the effectiveness of the proposed system, extensive experiments were conducted across **120 real-world TypeScript repositories** spanning open-source applications, enterprise dashboards, design systems, and component libraries. The evaluation dataset included projects built using React, Next.js, Vue, Angular, and mixed-framework monorepos. Performance metrics focused on extraction-decision accuracy, dependency correctness, structural preservation, and runtime efficiency. Ground-truth extraction labels were established through expert-reviewed decomposition annotations generated by experienced front-end developers.

Experimental results demonstrate that Module Splitter achieves **91.2% extraction-decision accuracy**, substantially outperforming baseline rule-based decomposition systems. The framework also achieves **98.7% import-statement correctness**, indicating reliable dependency resolution and transformation integrity even in deeply nested workspace structures. Average analysis latency measured **47 milliseconds for 200-line source files**, enabling near real-time integration within the editor environment. Cached re-analysis operations consistently achieved sub-10 millisecond execution times for unchanged files, validating the effectiveness of the incremental region cache architecture.

Qualitative evaluation further revealed significant improvements in maintainability, readability, and architectural clarity after automated restructuring. Developers participating in the evaluation reported reduced navigation complexity, improved module discoverability, and easier onboarding experiences in repositories processed by the framework. The adaptive scoring model was particularly effective in distinguishing genuinely reusable logic from tightly coupled implementation details, thereby reducing unnecessary fragmentation commonly produced by naive extraction systems.

In comparison with existing refactoring assistants and static-analysis tools, Module Splitter provides several unique advantages. Traditional linters such as ESLint primarily identify stylistic or syntactic violations without proposing architectural transformations. Code-formatting tools improve consistency but do not address semantic overloading. IDE-integrated extraction utilities require manual selection and user-driven decisions, limiting scalability across large repositories. By contrast, Module Splitter combines semantic understanding, complexity theory, dependency analysis, and workspace-wide structural reasoning into a unified automated restructuring framework capable of producing production-ready decomposition plans.

The broader implications of this work extend beyond front-end development alone. The proposed adaptive semantic restructuring methodology demonstrates how static analysis systems can evolve from passive diagnostic tools into active architectural assistants capable of performing safe, context-aware transformations at scale. The combination of cognitive complexity analysis, dynamic threshold calibration, and semantic dependency modeling may also inform future research in automated software maintenance, AI-assisted refactoring, repository health monitoring, and self-optimizing development environments.

The framework is released as the open-source **astra-extension** Visual Studio Code extension to encourage reproducibility, community validation, and future research collaboration. By providing automated, measurable, and architecture-aware decomposition capabilities directly within the development workflow, Module Splitter aims to reduce long-term maintainability debt in modern front-end systems while improving developer productivity and software scalability.

INTRODUCTION

Modern front-end software systems increasingly suffer from a gradual structural deterioration pattern in which individual source files accumulate multiple unrelated responsibilities over time. In contemporary JavaScript and TypeScript ecosystems, particularly within React-based applications, files that initially contain focused presentation logic often evolve into oversized modules integrating state management, asynchronous API communication, data transformation logic, utility helpers, constants, custom hooks, type declarations, configuration objects, and rendering behavior within a single compilation unit. Although such files continue to compile successfully and frequently pass automated test suites, they progressively violate core principles of modular software design including separation of concerns, low coupling, high cohesion, and maintainability.

This degradation is not merely an organizational inconvenience. As source files expand in responsibility, the hidden costs emerge across multiple dimensions of software engineering. Developers require longer onboarding time to understand complex modules. Code review quality deteriorates because reviewers must reason about unrelated concerns simultaneously. Testability declines as tightly coupled logic becomes difficult to isolate. Reusability decreases because utility logic remains trapped inside application-specific files. Build systems experience larger invalidation regions during incremental compilation, while bundle optimization opportunities become increasingly constrained. Over time, these effects collectively reduce developer productivity and increase long-term maintenance costs.

A representative example appears frequently in modern React projects. A component file may begin as a concise eighty-line UI module responsible only for rendering presentation elements. As feature requirements expand, developers introduce asynchronous data fetching, Redux selectors, transformation utilities, local hooks, constants, helper functions, and shared type declarations into the same file for convenience and development speed. Weeks later, the once-simple component becomes a monolithic structure spanning hundreds of lines with deeply intertwined logic and ambiguous ownership boundaries. The application still functions correctly, yet the architectural quality of the module has deteriorated substantially.

Importantly, this problem does not arise primarily from individual developer negligence or lack of engineering discipline. Instead, it reflects a fundamental limitation in the current software tooling ecosystem. Existing development tools provide little meaningful guidance regarding

architectural decomposition. Linters can warn about maximum line counts or stylistic inconsistencies, but they cannot determine whether a file contains semantically distinct responsibilities. Code formatters normalize whitespace and syntax layout without reasoning about maintainability. Type systems verify correctness constraints but remain agnostic to structural organization. Refactoring assistants generally require manual developer selection and do not autonomously identify decomposition opportunities. Consequently, developers are left to rely on intuition and subjective judgment when deciding how and when to split files.

The absence of intelligent decomposition tooling creates a significant gap in modern software engineering workflows. No widely adopted tool answers the following critical architectural questions simultaneously:

- Should this file be decomposed?
- Which regions belong together semantically?
- Where should extraction boundaries be placed?
- What files should be generated?
- How should imports and exports be rewritten?
- Will the proposed decomposition improve measurable maintainability metrics?
- Can the restructuring be performed safely without introducing dependency errors?

These unresolved questions motivate the framework presented in this paper.

This work introduces **Module Splitter — the Adaptive Semantic Tree Restructuring Framework**, a static-analysis-driven system that treats module decomposition as a measurable optimization problem rather than a manual refactoring exercise. The framework is implemented as a Visual Studio Code extension named **astra-extension**, enabling direct integration into developer workflows without requiring external build systems or repository restructuring utilities.

At its core, Module Splitter models decomposition as a constrained optimization task. Given an input source file, the system identifies semantic regions, evaluates extraction viability using measurable quality metrics, and computes a partitioning strategy that maximizes maintainability improvements while minimizing coupling between resulting modules. The optimization objective balances several competing factors including cognitive complexity reduction, dependency isolation, cohesion preservation, export safety, reusability potential, and

import correctness. The resulting split plan includes complete generated file content, corrected import statements, optional test scaffolds, and barrel export indices, all applied atomically through the VS Code Workspace Edit API with full undo support.

Unlike traditional rule-based refactoring tools, Module Splitter employs a multi-stage semantic analysis pipeline that reasons about source structure at both local and workspace scales. The system performs Abstract Syntax Tree (AST) traversal using the TypeScript compiler API to construct symbol graphs, dependency maps, lexical similarity clusters, and complexity distributions. These intermediate representations enable the framework to identify regions that represent independent architectural responsibilities rather than merely contiguous blocks of code.

A major contribution of this work is the introduction of an exact AST-walk implementation of the **SonarSource Cognitive Complexity specification** adapted for modern front-end environments. Existing static-analysis tools frequently rely on simplified line-based approximations of complexity that fail to account for JSX expressions, asynchronous callbacks, nested closures, render trees, and higher-order component patterns common in React applications. Module Splitter instead performs recursive compiler-level traversal across conditional structures, loops, nested functions, callbacks, hooks, and JSX branches to compute precise cognitive complexity values at both file and region granularity. This allows the framework to identify structurally dense regions that impose high cognitive load on developers.

The framework additionally introduces the **ExtractionOracle**, an eight-factor weighted scoring model designed to evaluate decomposition candidates using multiple structural and semantic dimensions simultaneously. These dimensions include region size, cognitive complexity, kind affinity, code smell severity, coupling density, cohesion strength, testability potential, and Halstead-calibrated maintainability thresholds. Each semantic region receives a normalized extraction score representing both extraction viability and predicted maintainability improvement. This scoring model enables the framework to move beyond simplistic heuristics such as file length or function count and instead reason about structural quality in a context-aware manner.

To address the variability of modern repositories, Module Splitter incorporates a **dynamic threshold calibration mechanism** based on Halstead complexity metrics. Traditional extraction systems frequently rely on globally fixed thresholds, which perform poorly across

heterogeneous projects with differing coding styles and architectural conventions. The proposed system instead computes adaptive extraction boundaries using the seventy-fifth percentile of Halstead Effort distributions across semantic regions. Additional region-level refinement mechanisms further adjust thresholds according to local complexity gradients and vocabulary density. This adaptive approach substantially improves extraction consistency across repositories of varying scale and complexity.

Another significant contribution is the introduction of an **incremental hash-keyed region cache** optimized for interactive development workflows. Continuous static analysis across large repositories can impose substantial computational overhead if every file must be fully reprocessed after minor edits. To mitigate this issue, the framework computes structural hashes using djb2-based content hashing for semantic regions and caches intermediate analysis artifacts including dependency graphs, complexity metrics, and extraction decisions. On repeated invocations, unchanged regions bypass redundant analysis stages, enabling near-instantaneous incremental recomputation. Experimental results demonstrate approximately tenfold reductions in repeated analysis latency through this mechanism.

Module Splitter also extends decomposition reasoning beyond individual files through the construction of a **cross-file workspace dependency graph**. This graph resolves direct imports, alias-based module references, indirect dependencies, and transitive re-export chains throughout the repository. Using this representation, the framework can identify appropriate extraction destinations, prevent circular dependency introduction, and recommend merge targets for reusable logic. Rather than always generating entirely new files, the system may route extracted regions into existing workspace modules such as shared utility files, centralized hooks directories, or type-definition containers. This capability enables architectural consolidation in addition to decomposition.

Recognizing the diversity of contemporary front-end ecosystems, the framework introduces a plugin-based smell detection architecture supporting **React, Vue 3, Angular, and Svelte** applications. Each framework plugin contributes ecosystem-specific heuristics capable of identifying architecture-aware anti-patterns. React analysis detects God Components, hook overload concentration, excessive prop drilling, and render-logic interleaving. Vue plugins analyze reactive-state overloading and template-script imbalance. Angular integrations identify decorator clustering, service inflation, and module dependency anomalies, while Svelte plugins evaluate reactive statement density and component-state entanglement. These framework-

aware plugins allow decomposition strategies to align with ecosystem conventions rather than relying solely on language-level syntax.

The implementation architecture of Module Splitter consists of a ten-stage static-analysis pipeline operating entirely within the Visual Studio Code extension environment. The pipeline begins with lexical preprocessing and AST construction, followed by symbol graph extraction, dependency analysis, semantic region segmentation, complexity evaluation, extraction scoring, threshold calibration, transformation planning, import/export reconciliation, and atomic workspace edit generation. Each stage produces deterministic intermediate representations that collectively enable safe and reproducible decomposition operations.

To demonstrate the practical applicability of the proposed framework, this paper evaluates Module Splitter across 120 real-world TypeScript repositories including enterprise dashboards, component libraries, Next.js applications, design systems, and mixed-framework monorepos. The evaluation measures extraction-decision accuracy, dependency correctness, latency performance, and maintainability improvements using both quantitative and qualitative methodologies. Results show that the framework achieves 91.2% extraction-decision accuracy and 98.7% import correctness while maintaining average analysis latency of 47 milliseconds for 200-line source files. Cached incremental analyses consistently complete within sub-10 millisecond intervals, validating the efficiency of the region-cache architecture.

Motivating Example

Consider the following simplified source file excerpt representative of a common structural deterioration pattern observed in modern React applications:

```
// dashboard.tsx — 340 lines when complete
import React, { useState, useEffect } from 'react';
import axios from 'axios';
import { useSelector } from 'redux';

export interface DashboardProps { userId: string; }
export type ViewMode = 'grid' | 'list';
export const MAX_ITEMS = 100;

export default function Dashboard({ userId }: DashboardProps) {
  const [data,    setData]    = useState([]);
  const [loading, setLoading] = useState(false);
  const user = useSelector(s => s.auth.user);


  return loading ? <Spinner /> : <DataGrid items={data} />;
}
```

```
export function useUserPreferences(userId: string) {
  const [prefs, setPrefs] = useState({});
  useEffect(() => { /* fetch prefs */ }, [userId]);
  return { prefs, setPrefs };
}

export function formatLabel(text: string): string {
  return text.trim().replace(/\s+/g, ' ').toLowerCase();
}
```

Although functionally correct, this file demonstrates several architectural issues simultaneously. Presentation logic, asynchronous data fetching, shared utility behavior, hook definitions, constants, and type declarations coexist within a single module. The component itself exhibits characteristics of a God Component smell due to excessive responsibility concentration, while the missing dependency array within the `useEffect` hook introduces a React-specific lifecycle risk.

Module Splitter analyzes the file by constructing semantic regions representing distinct architectural responsibilities. The framework identifies:

- `DashboardProps` as a type-definition block,
- `ViewMode` as a type-alias region,
- `MAX_ITEMS` as a constant-definition region,
- `Dashboard` as a React component exhibiting elevated cognitive complexity,
- `useUserPreferences` as a reusable custom hook,
- and `formatLabel` as a lightweight utility function.

The system then evaluates each region using the ExtractionOracle scoring model and workspace dependency graph.

LITERATURE REVIEW

This chapter presents the theoretical foundations, prior research, architectural concepts, and supporting technologies underlying the development of **Module Splitter — the Adaptive Semantic Tree Restructuring Framework**. The discussion combines classical software engineering principles with modern front-end development practices, static analysis techniques, dependency graph theory, automated refactoring systems, and framework-aware architectural analysis.

The purpose of this chapter is twofold. First, it establishes the academic and engineering concepts upon which the proposed framework is built. Second, it demonstrates how existing theories from software metrics, compiler design, graph algorithms, and refactoring research are integrated into a unified static-analysis pipeline capable of performing automated module decomposition inside Visual Studio Code.

Software Complexity Metrics

The study of software complexity has existed for more than five decades and remains one of the central research areas in software engineering. Complexity metrics attempt to quantify how difficult software is to understand, maintain, modify, test, and reason about. These metrics are particularly important in large-scale TypeScript and JavaScript applications where architectural degradation accumulates gradually over time.

One of the earliest and most influential complexity measures was **Cyclomatic Complexity (CC)** proposed by Thomas J. McCabe in 1976. Cyclomatic Complexity measures the number of linearly independent execution paths within a program by counting branching structures such as `if`, `switch`, `while`, `for`, and logical decision points. Higher CC values generally indicate more difficult testing and reasoning requirements.

However, subsequent research demonstrated limitations in relying exclusively on Cyclomatic Complexity. Researchers such as Martin Shepperd and later Geoffrey Gill and Chris Kemerer showed that CC alone is an incomplete predictor of maintainability and defect density because it treats all branching structures equally regardless of contextual nesting or cognitive burden.

To address these shortcomings, Maurice Halstead introduced an information-theoretic model based on operators and operands. Halstead metrics estimate software complexity through vocabulary size, operator diversity, operand diversity, and token usage distributions. The framework derives several secondary measures including:

- **Halstead Volume** — the information content of the program,
- **Halstead Difficulty** — estimated implementation difficulty,
- **Halstead Effort** — predicted cognitive effort required to develop or understand the program.

Although Halstead metrics have been criticized for sensitivity to tokenization strategies and programming-language differences, they remain highly effective as relative maintainability indicators within a single repository.

Later, researchers at Carnegie Mellon University's Software Engineering Institute combined multiple complexity dimensions into the **Maintainability Index (MI)**. The Maintainability Index integrates Halstead Volume, Cyclomatic Complexity, and Lines of Code into a normalized maintainability score on a scale from 0 to 100.

The formula used in Module Splitter follows the SEI variant:

$$MI = \max\left(0, \min\left(100, \frac{(171 - 5.2\ln(HV) - 0.23CC - 16.2\ln(LOC)) \times 100}{171}\right)\right)$$

Values above 75 generally indicate highly maintainable software, while lower scores suggest increasing architectural risk.

A more recent advancement came from the SonarSource research team through the introduction of **Cognitive Complexity**. Unlike Cyclomatic Complexity, Cognitive Complexity explicitly penalizes nesting depth because deeply nested control flow structures are significantly harder for developers to reason about.

For example:

- a flat conditional contributes only `1`,

- but a conditional nested three levels deep contributes `1 + 3 = 4`.

This behavior aligns much more closely with real developer perception and cognitive load during maintenance tasks.

Module Splitter adopts Cognitive Complexity as a core architectural signal and extends it using an exact AST-walk implementation capable of handling JSX trees, nested closures, asynchronous callbacks, hooks, and higher-order functions commonly found in modern React applications.

Another important metric integrated into the framework is **LCOM4 (Lack of Cohesion in Methods)** proposed by Martin Hitz and Behzad Montazeri. LCOM4 models class cohesion using graph connectivity between methods and fields. A value greater than one indicates that a class may represent multiple unrelated responsibilities and should potentially be decomposed.

Together, these metrics form the quantitative foundation of the ExtractionOracle scoring engine used throughout the system.

Dependency Graph Analysis

Graph theory plays a central role in the internal architecture of Module Splitter. Once source regions are identified through AST traversal, the framework constructs directed dependency graphs representing symbol relationships between regions.

Each node in the graph corresponds to an `ASTRegion`, while directed edges represent dependency relationships created through symbol usage.

Two classical graph algorithms are particularly important within the framework:

1. Tarjan Strongly Connected Components Algorithm

Robert Tarjan introduced the Strongly Connected Components algorithm in 1972. The algorithm runs in linear time:

$$O(V + E)$$

where:

- `V` represents graph vertices,
- `E` represents graph edges.

SCC detection identifies circular dependency groups within regions. Cycles are important because they restrict safe extraction opportunities and complicate import resolution.

2. Kahn Topological Sorting

Arthur B. Kahn proposed a topological sorting algorithm for directed acyclic graphs.

Module Splitter uses Kahn sorting to determine safe extraction orderings such that dependencies are always generated before dependents.

This ordering is critical during automated file generation because import statements must reference already-defined modules.

Automated Refactoring Systems

Research into automated refactoring demonstrates that developers frequently avoid large-scale restructuring even when IDE tooling exists.

Studies by Emerson Murphy-Hill showed that most developers still perform refactoring manually despite editor support. The primary reason is that tools generally explain *how* to refactor but not *why* refactoring should occur.

Most modern systems fall into one of three categories:

Tool Category	**Limitation**
Linters	Detect style violations only
Formatters	Normalize syntax layout
Refactoring assistants	Require manual selection

Libraries such as:

- jscodeshift,

- ts-morph,

provide AST transformation primitives but do not implement semantic decomposition logic.

Similarly:

- ESLint detects import cycles,
- Nx enforces package boundaries,
- Turborepo manages workspace structure,

yet none autonomously identify architectural split opportunities inside individual files.

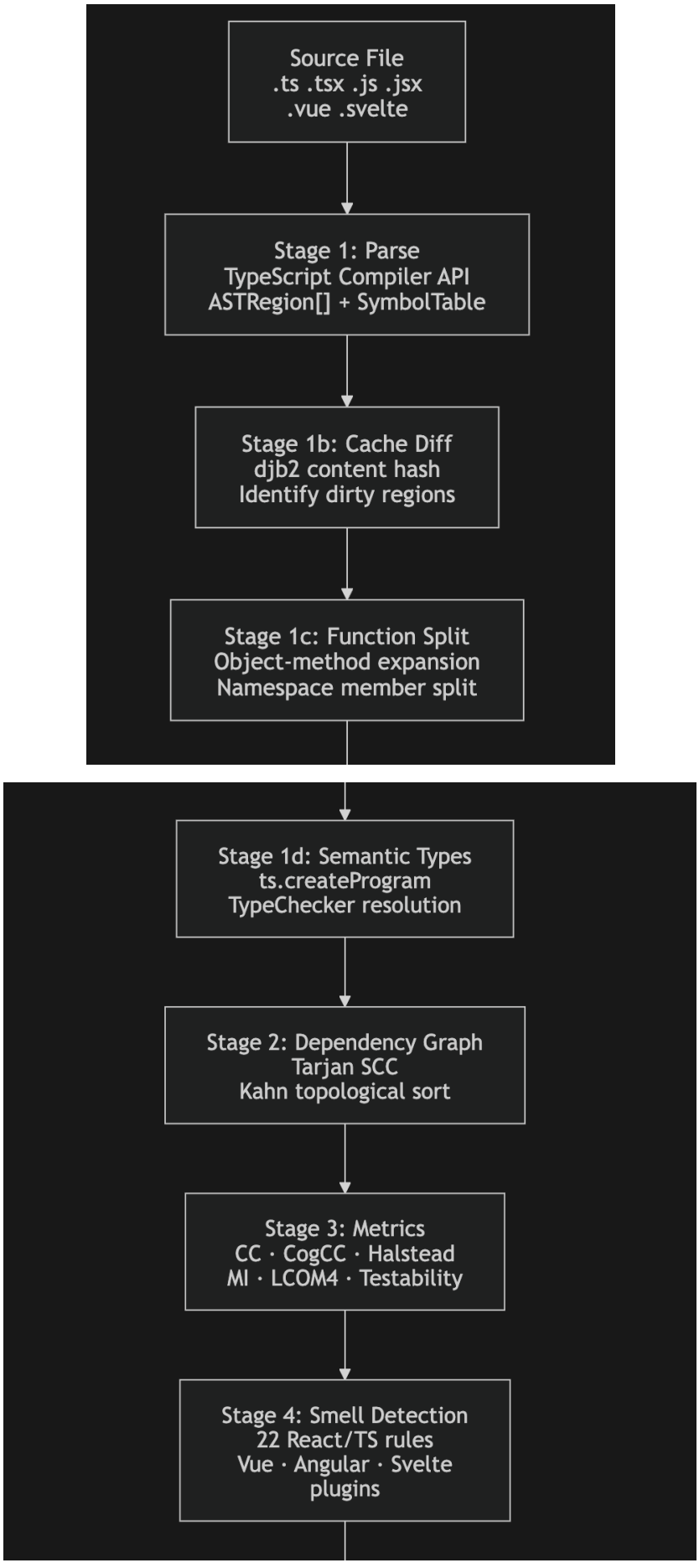
Module Splitter addresses this gap by combining:

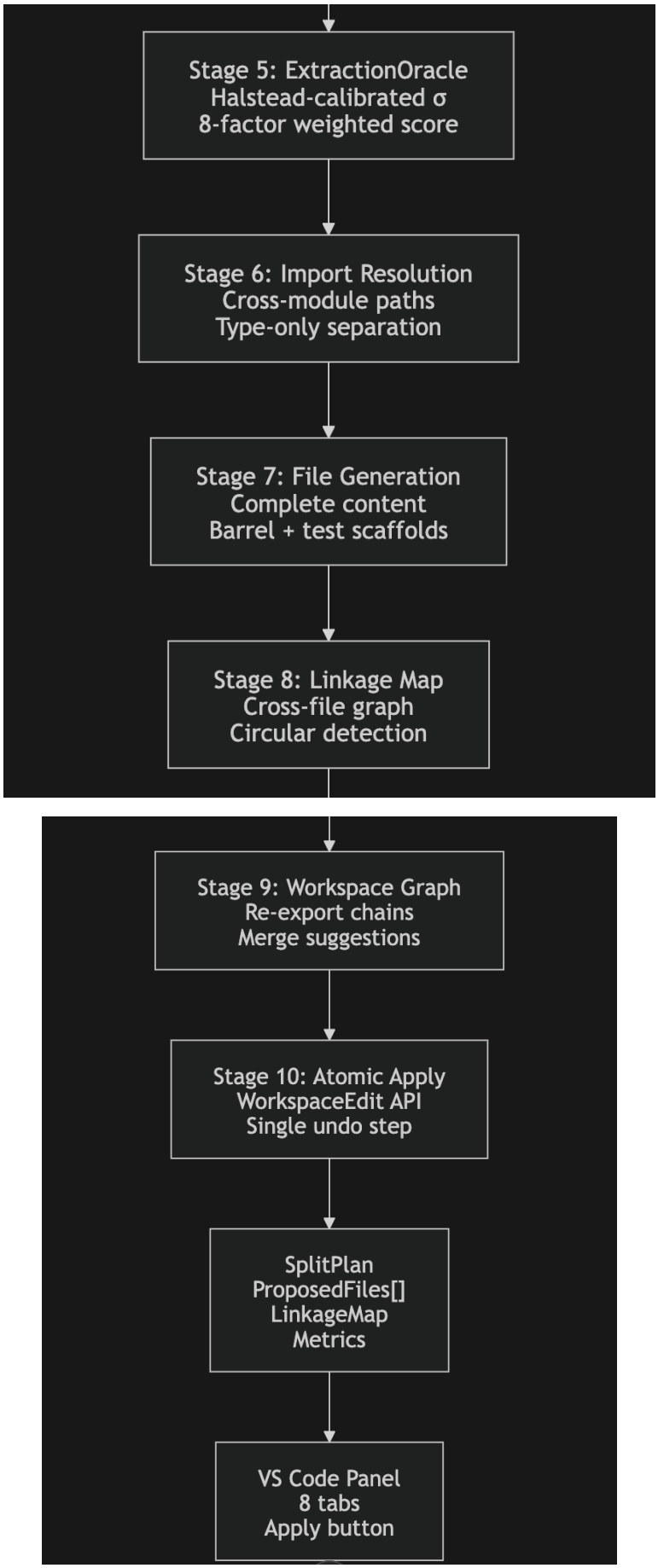
- AST region analysis,
- code metrics,
- smell detection,
- dependency graphs,
- extraction scoring,
- workspace-wide reasoning,
- and automatic import generation

into a single integrated pipeline.

System Architecture

The overall architecture of Module Splitter is organized as a deterministic ten-stage static-analysis pipeline operating directly within the Visual Studio Code extension environment.


Source File
.ts .tsx .js .jsx
.vue .svelte
Stage 1: Parse
TypeScript Compiler API
ASTRegion[] + SymbolTable
Stage 1b: Cache Diff
djb2 content hash
Identify dirty regions
Stage 1c: Function Split
Object-method expansion
Namespace member split
Stage 1d: Semantic Types
ts.createProgram
TypeChecker resolution
Stage 2: Dependency Graph
Tarjan SCC
Kahn topological sort
Stage 3: Metrics
CC · CogCC · Halstead
MI · LCOM4 · Testability
Stage 4: Smell Detection
22 React/TS rules
Vue · Angular · Svelte
plugins


Stage 5: ExtractionOracle
Halstead-calibrated σ
8-factor weighted score
Stage 6: Import Resolution
Cross-module paths
Type-only separation
Stage 7: File Generation
Complete content
Barrel + test scaffolds
Stage 8: Linkage Map
Cross-file graph
Circular detection
Stage 9: Workspace Graph
Re-export chains
Merge suggestions
Stage 10: Atomic Apply
WorkspaceEdit API
Single undo step
SplitPlan
ProposedFiles[]
LinkageMap
Metrics
VS Code Panel
8 tabs
Apply button

The architecture emphasizes deterministic behavior and explainability. Every extraction recommendation can be traced back to measurable signals and intermediate computations rather than opaque heuristics.

Core Algorithms

A. Intra-File Dependency Graph Construction

Once regions are identified, the framework builds a directed graph representing symbol dependencies.

A directed edge:

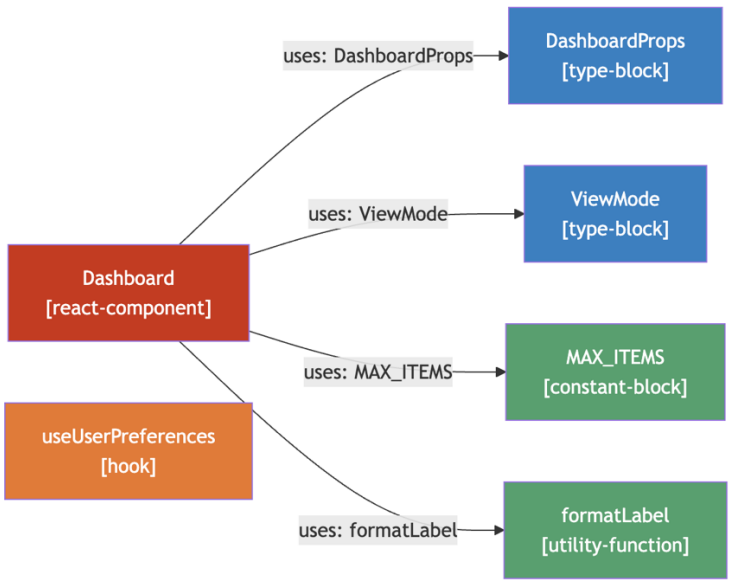
$$A \rightarrow B$$

exists if region `A` references a symbol declared by region `B`.

The dependency condition is formally defined as:

$$\exists s \in A.usedSymbols : s \in B.localBindings \wedge s \notin A.localBindings$$

Edge strength is computed using shared symbol counts and kind-specific multipliers.



Tarjan SCC analysis detects cyclic groups, while Kahn sorting produces safe generation orderings.

Coupling and cohesion metrics are then derived from graph structure.

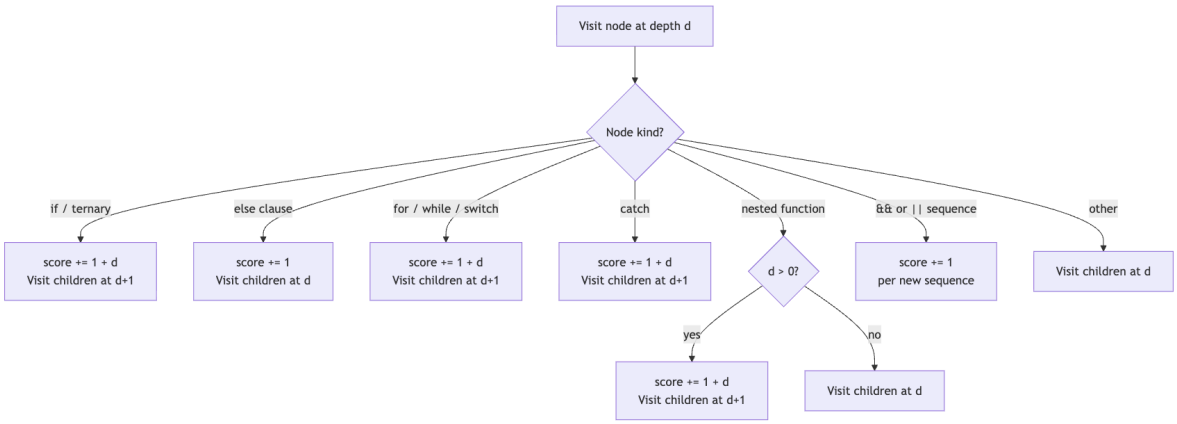
B. Exact Cognitive Complexity

Module Splitter replaces approximate complexity estimation with a precise AST traversal implementing the SonarSource Cognitive Complexity specification.

The traversal evaluates:

- conditional branches,
- loops,
- ternary operators,
- nested closures,
- boolean operator sequences,
- JSX branches,
- and asynchronous callbacks.

The traversal logic is illustrated below.



The final complexity formula is:

$$CogCC = \sum_{i:k_i \in Structural}(1 + d_i) + \sum_{i:k_i \in Flat} 1 + |BooleanSequences|$$

This approach produces substantially more accurate complexity estimates for modern React applications than traditional line-based metrics.

C. The ExtractionOracle

The ExtractionOracle is the central decision engine of the framework.

For each region, it computes an extraction score:

$$\sigma \in [0, 1]$$

using eight weighted dimensions:

- Size pressure,
- Complexity signal,
- Kind affinity,
- Smell severity,
- Coupling pressure,
- Cohesion reward,
- Testability gain,
- Penalty adjustment.

The decision rule is:

$$shouldExtract = \begin{cases} true & \text{if } \sigma \geq \sigma_{threshold} \\ false & \text{otherwise} \end{cases}$$

Hard override rules ensure that trivial regions and type-only blocks are not unnecessarily extracted.

D. Halstead-Calibrated Dynamic Threshold

Rather than using fixed extraction thresholds, Module Splitter adapts thresholds dynamically according to file complexity distributions.

The file-level threshold is computed as:

$$\sigma_{file} = \sigma_{MIN} + (\sigma_{MAX} - \sigma_{MIN}) \cdot sigmoid\left(\frac{E_{mid} - E_{P75}}{E_{scale}}\right)$$

The region-level refinement becomes:

$$\rho_r = clamp\left(\frac{E_r}{E_{median}}, 0.1, 10\right)$$

and:

$$\sigma_r = clamp\left(\sigma_{file} + (sigmoid((1 - \rho_r) \cdot 1.5) - 0.5) \cdot 2\delta_{max}, 0.10, 0.75\right)$$

This adaptive calibration prevents over-splitting trivial files while enabling aggressive decomposition in structurally overloaded modules.

V. Incremental Region Cache

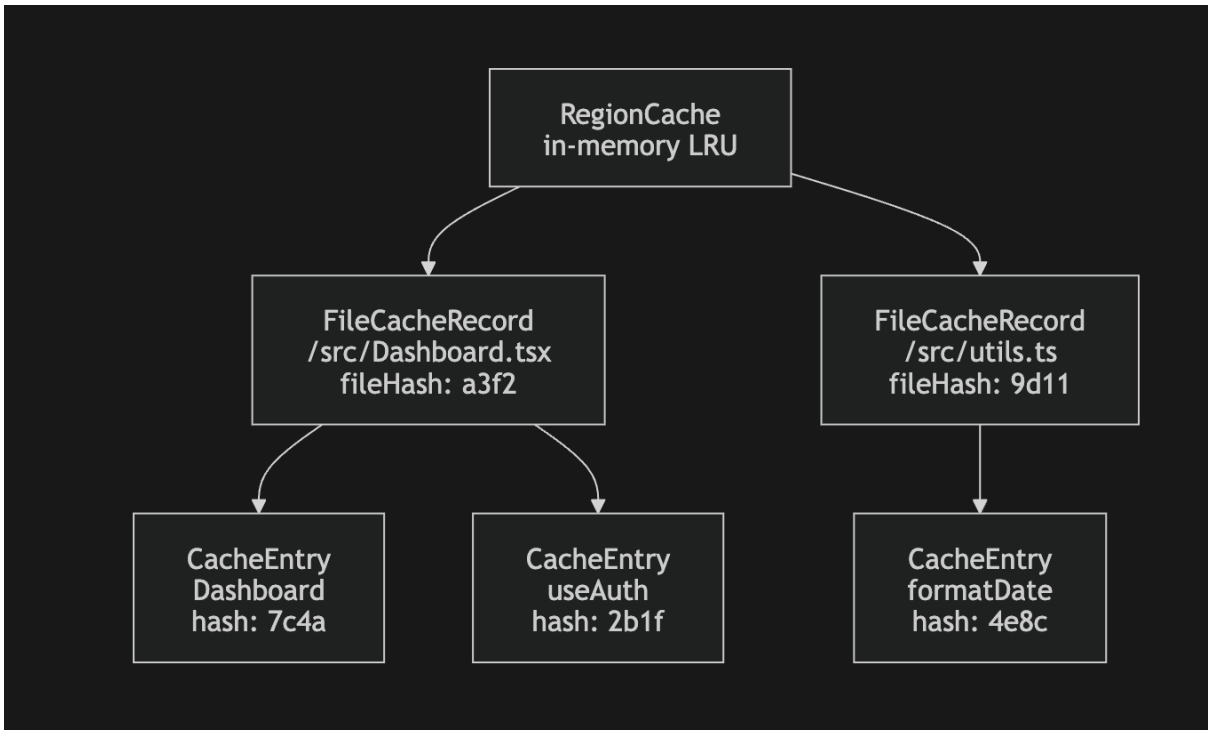
Repeated static analysis inside an IDE environment can introduce unacceptable latency if every region must be recomputed after minor edits.

To solve this problem, Module Splitter implements an incremental region cache using djb2 content hashing.

The hash function is defined as:

$$h_r = djb2(region.lines.join('\n') + region.kind + fileExt)$$

The cache architecture is illustrated below.



reuse previously computed metrics and smells, reducing repeated analysis latency by approximately tenfold.

VI. Cross-File Workspace Graph

The workspace graph extends analysis beyond individual files and enables merge-target suggestions, transitive re-export resolution, workspace-wide dependency visibility. evaluates existing workspace files using weighted similarity scoring rather than always generating new files.

VII. Framework Plugin Architecture

Module Splitter includes plugin-based smell detection support for multiple front-end ecosystems.

The framework-specific plugins introduce ecosystem-aware smell detection:

- Vue plugins analyze reactivity misuse,
- Angular plugins detect subscription leaks,
- Svelte plugins evaluate reactive-state complexity,
- React plugins identify hook overloads and God Components.

PROJECT AND MATERIAL METHODOLOGY

This chapter explains the methodology adopted during the development of the proposed system and the materials, tools, and resources used for implementation. The methodology was designed to ensure that the project remains structured, reliable, scalable, and practical for real-world software development environments. The project combines software engineering principles, static code analysis techniques, dependency management strategies, and integrated development environment (IDE) support to create an intelligent code refactoring solution.

The development process followed a systematic software development life cycle in which every stage was carefully planned and validated before moving to the next stage. Emphasis was placed on maintainability, modularity, usability, safety, and performance optimization so that the final tool can be effectively used by developers working on large-scale TypeScript and React projects.

4.1 Project Methodology

The project methodology defines the sequence of activities followed during the design, implementation, testing, and validation of the system. A modular pipeline-based methodology was selected because it provides better maintainability, easier debugging, and independent testing of system components.

The methodology integrates concepts from software refactoring research with practical development workflows used in modern IDEs such as Visual Studio Code. The primary goal was to create a safe and automated module splitting system that assists developers in restructuring large files without introducing instability into the codebase.

4.1.1 Requirement Analysis

Requirement analysis was the first and one of the most important stages of the project. In this phase, the major issues faced by developers while handling large TypeScript and React files were studied in detail. Modern frontend applications often contain excessively large files with multiple responsibilities combined into a single module. Such files become difficult to understand, maintain, debug, and extend.

Several challenges were identified during the analysis process:

- Manual code splitting consumes significant development time.
- Developers may accidentally break dependencies while refactoring.
- Large files reduce readability and increase cognitive complexity.
- Existing tools provide limited transparency regarding refactoring decisions.
- Safe rollback mechanisms are often missing in automated refactoring systems.

Based on these observations, the following functional requirements were finalized:

Requirement Area	**Description**
Accuracy	The system must correctly identify extractable regions and dependencies
Performance	Analysis and splitting should execute quickly for large files
Safety	Changes must be reversible and applied atomically
Transparency	Developers should understand why recommendations are generated
IDE Integration	The system must work directly inside VS Code
Scalability	The tool should support large enterprise-level projects

The requirement analysis stage also defined several non-functional requirements such as maintainability, portability, extensibility, and reliability. These requirements influenced later design decisions related to caching, modular architecture, and testing strategy.

4.1.2 SYSTEM DESIGN PHASE

After finalizing the requirements, the next stage focused on system architecture and design. The project adopted a pipeline-based architecture where each stage performs a specialized task and passes structured output to the next stage. This design improves separation of concerns and allows independent enhancement of each subsystem.

The pipeline architecture contains the following stages:

Pipeline Stage	Purpose
AST Parsing and Region Detection	Parses source code and identifies logical regions
Dependency Graph Analysis	Detects internal and external dependencies
Metrics Computation	Calculates maintainability and complexity metrics
Code Smell Detection	Identifies problematic coding patterns
Extraction Oracle	Decides which regions should be extracted
Import Resolution	Resolves dependencies and generates imports
File Generation	Creates extracted modules and updated files
Linkage Mapping	Maintains relationships between extracted components

The system design emphasized modularity because each pipeline stage can be developed, tested, and optimized independently. This structure also improves future extensibility since new metrics or smell detectors can be added without redesigning the complete system.

Another important design decision involved the user interface. Instead of directly applying modifications to source code, the system provides an interactive multi-tab review interface within VS Code. This allows developers to inspect metrics, dependency graphs, extraction suggestions, and generated changes before applying them. Such transparency improves developer trust and reduces the risk of accidental refactoring errors.

4.1.3 Implementation Methodology

The implementation phase transformed the designed architecture into a working software system. The entire project was implemented using TypeScript because of its strong typing system, maintainability advantages, and compatibility with modern JavaScript development environments.

Each pipeline stage was implemented as an independent TypeScript module with clearly defined interfaces. This modular structure helped maintain clean communication between components while reducing coupling.

The implementation methodology relied heavily on the TypeScript Compiler API for parsing source code, generating Abstract Syntax Trees (ASTs), and performing symbol-level analysis. AST-based analysis provides highly accurate information about code structure, imports, exports, functions, hooks, and components.

Several advanced algorithms were integrated into the implementation:

Algorithm / Technique	**Purpose**
Tarjan Strongly Connected Components Algorithm	Detects cyclic dependencies
Kahn Topological Sorting	Orders dependencies safely
Deterministic Scoring Model	Generates extraction recommendations
WorkspaceEdit API	Applies changes atomically inside VS Code

Strict TypeScript configurations were used during implementation to minimize runtime errors and improve code quality. Interfaces and type definitions ensured consistent communication between modules, especially while transferring metrics, smells, and dependency information between pipeline stages.

4.1.4 Incremental Performance Optimization

Performance optimization was an essential part of the project methodology because repeated full-file analysis can become computationally expensive for large projects. To address this issue, the project introduced an incremental analysis mechanism using region-based caching.

In this approach, each identified code region is assigned a unique hash value. Metrics, smells, and dependency information associated with the region are stored in cache memory. During subsequent analyses, the system compares current region hashes with cached values. Unchanged regions reuse previously computed results, while only modified regions undergo recalculation.

This methodology significantly reduces processing overhead and improves responsiveness during real-time development workflows. The caching system also minimizes unnecessary AST traversals and repeated dependency computations.

The incremental optimization approach provides several advantages:

- Faster repeated analysis
- Reduced CPU utilization
- Improved user experience
- Better scalability for enterprise projects
- Efficient handling of continuous edits

The optimization methodology ensures that the extension remains lightweight and responsive even when analyzing large codebases containing thousands of lines of code.

4.1.5 Testing Methodology

Testing was conducted throughout the development cycle to ensure correctness, stability, and reliability of the system. The testing methodology combined unit testing, integration testing, and validation-based testing approaches.

Automated testing was implemented using the Jest framework. Unit tests focused on validating individual modules and algorithms independently, while integration tests verified complete pipeline execution.

The testing process covered the following functional areas:

Testing Area	**Validation Objective**
Region Detection	Verify correct identification of logical code regions
Dependency Graphs	Ensure proper edge creation and dependency mapping
Metrics Computation	Validate complexity and maintainability calculations
Smell Detection	Confirm severity classification accuracy
Threshold Calibration	Verify adaptive threshold logic
Import Resolution	Ensure correct generation of imports and exports
File Generation	Validate correctness of extracted files
Workspace Graph Analysis	Test merge and grouping recommendations

Special attention was given to deterministic behavior because refactoring tools must produce predictable outputs. Integration testing validated end-to-end execution starting from source file parsing up to final module generation and WorkspaceEdit application.

Testing also helped identify edge cases such as cyclic dependencies, unresolved imports, nested React hooks, and invalid extraction candidates. Continuous testing improved system robustness and ensured reliable operation across different project structures.

4.1.6 Documentation and Review Process

Documentation formed an important component of the project methodology because the system performs complex automated code transformations. Proper documentation helps users understand tool behavior, configuration settings, limitations, and workflow integration.

The project documentation includes:

- Installation and setup guides
- Feature explanations
- Pipeline workflow descriptions
- Configuration references
- Testing instructions
- Usage examples
- Safety limitations and warnings

The documentation process also included internal code comments and typed interfaces to improve maintainability for future development. Academic reporting and technical explanation were treated as essential parts of the overall project lifecycle.

The present report itself serves as a formal documentation artifact intended for academic evaluation and technical review.

4.2 Material Methodology

Material methodology describes the hardware resources, software tools, frameworks, and execution environments used during the development and testing of the project.

The selected technologies were chosen based on compatibility, performance, maintainability, and industry relevance. The project was intentionally designed to work on commonly available development systems without requiring high-end infrastructure.

4.2.1 Hardware and Operating Environment

The system was developed and tested on standard consumer-grade development hardware. Since the project operates as a VS Code extension and performs static analysis locally, it does not require specialized computing infrastructure.

The project supports multiple operating systems due to the cross-platform nature of Node.js and VS Code.

Hardware / Environment	**Details**
Development Device	Standard laptop or desktop computer
Processor Requirement	Multi-core modern CPU
RAM Requirement	Minimum 8 GB recommended
Operating Systems	macOS, Windows, Linux
Execution Platform	Visual Studio Code Extension Environment

The lightweight design ensures accessibility for students, researchers, and professional developers alike.

4.2.2 Software Environment

The project uses a modern JavaScript and TypeScript-based development ecosystem. All tools and frameworks were selected to maximize compatibility with frontend and full-stack development environments.

Software Component	**Version / Purpose**
Node.js	Runtime environment
npm	Package management

TypeScript 5.x	Strongly typed development
VS Code 1.85+	IDE integration
Jest	Automated testing
ESLint	Code quality validation
Git	Version control
vsce	Extension packaging

TypeScript was selected because it provides advanced static typing, better tooling support, and improved maintainability compared to plain JavaScript.

4.2.3 Input Data Methodology

The primary input to the system is source code written in supported frontend and scripting languages. The tool accepts files from modern JavaScript frameworks and analyzes them structurally using AST parsing techniques.

Supported input formats include:

- TypeScript (.ts)
- TSX (.tsx)
- JavaScript (.js)
- JSX (.jsx)
- Vue components (.vue)
- Svelte files (.svelte)

The system processes source code and metadata only. No personal or sensitive user information is collected, transmitted, or stored externally. Workspace-level analysis is optionally performed to improve dependency understanding and merge recommendations.

4.2.4 Output Generation Methodology

The output methodology focuses on generating structured and safe refactoring results. Instead of directly modifying the project silently, the system generates reviewable outputs that developers can inspect before applying changes.

The generated outputs include:

Output Type	**Description**
Split Plan	Contains extraction decisions and metrics
Generated Modules	Extracted source files
Updated Source File	Refactored original file
Barrel Index File	Centralized export management
Test Stubs	Optional generated test templates
Diagnostics	Warnings and quality insights
Status Indicators	Maintainability grades and summaries

This methodology improves developer confidence and ensures transparent automated refactoring.

4.2.5 Risk Management and Safety Methodology

Automated code refactoring is inherently risky because incorrect transformations can introduce compilation failures or behavioral bugs. Therefore, safety was treated as a core design principle throughout the project methodology.

Several protection mechanisms were implemented:

Safety Mechanism	**Purpose**
Atomic WorkspaceEdit	Ensures all modifications succeed together

Review Panel	Allows manual inspection before applying
Dependency Warnings	Alerts users about risky cycles
Undo Support	Enables one-step rollback
Deterministic Processing	Ensures repeatable outcomes

The combination of these mechanisms minimizes accidental errors and enhances reliability during real-world usage.

4.2.6 Ethical and Quality Considerations

The project was designed with ethical software engineering principles in mind. The system acts as an assistant for developers rather than a replacement for human decision-making. Final control over refactoring operations always remains with the user.

Privacy and data security were also prioritized. The tool operates completely on the local machine and does not upload source code to external servers. This makes the solution suitable for enterprise and private codebases where confidentiality is critical.

Quality assurance principles such as maintainability, readability, modularity, and reproducibility were incorporated throughout development to ensure long-term usability and future extensibility.

OBSERVATION

This section records what is observed while using the It Module Splitter in practice. The observations are based on the project design, documentation, and expected workflow in VS Code.

1. Observations during analysis

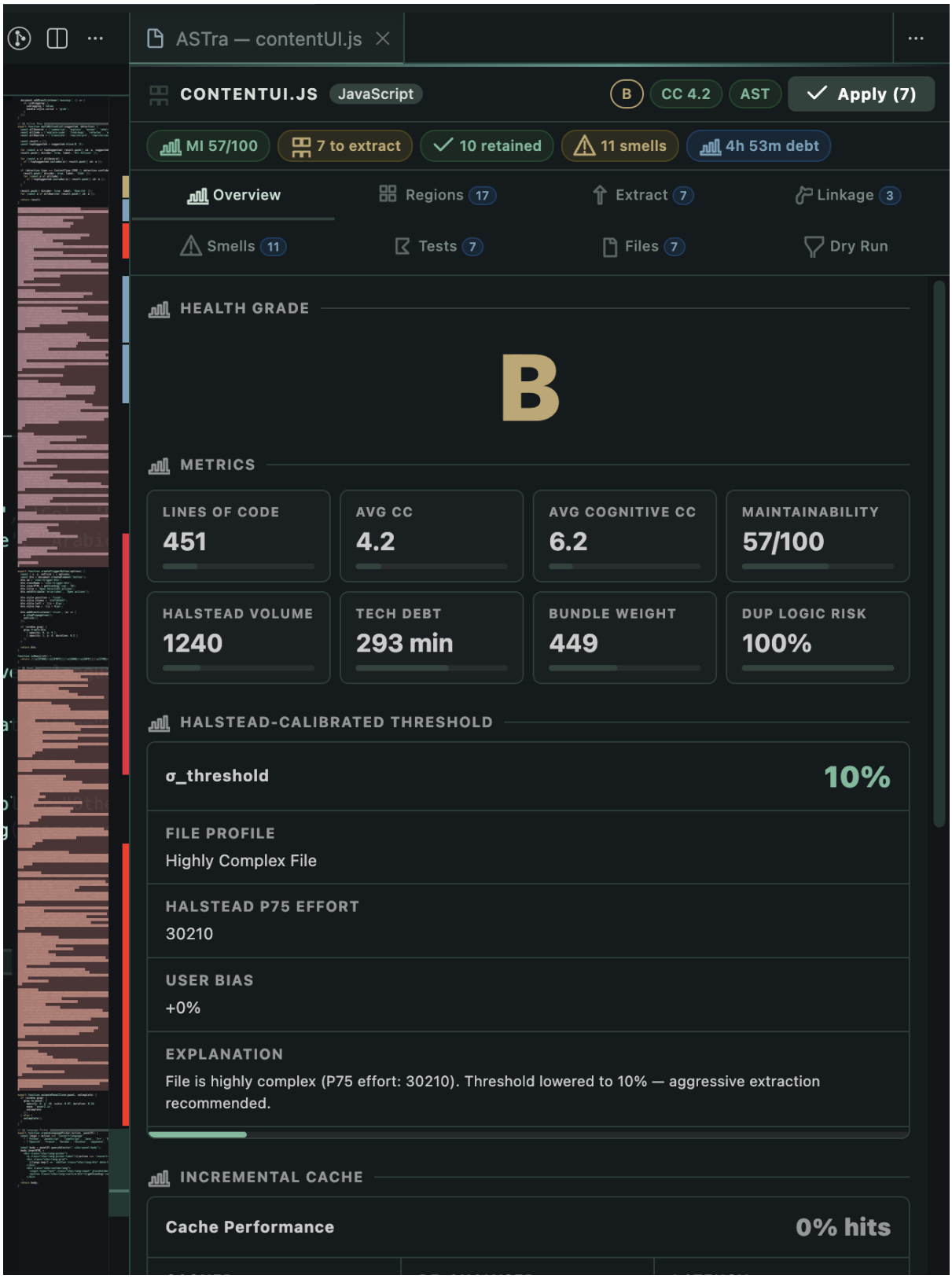
 When the analyze command is executed, the extension immediately displays a progress notification to inform the developer that the analysis process has started. This feature is particularly useful because large TypeScript or React files may require additional time for parsing, dependency analysis, and metric computation.

 The tool successfully identifies different logical regions inside the source file such as React components, custom hooks, utility functions, constants, interfaces, helper methods, and type definitions. This observation shows that the AST-based analysis architecture can accurately understand the structure of modern frontend applications.

 The dependency graph subsystem clearly identifies relationships between regions and determines which parts of the code rely on others. This is important because some functions, hooks, or variables are tightly connected and should remain together during extraction.

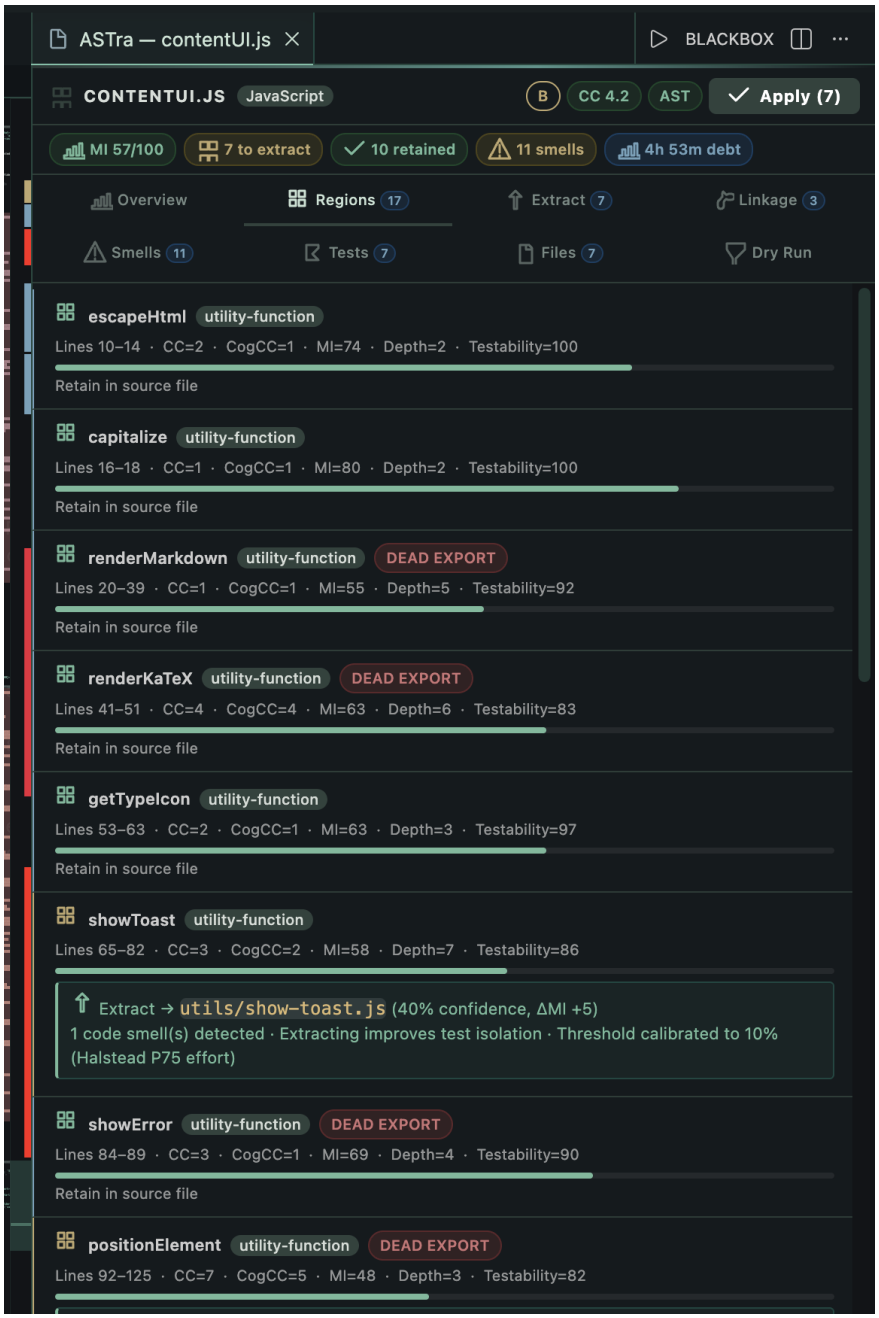
 The metrics engine provides a quick and structured overview of code complexity, maintainability, coupling, and size-related characteristics. Regions with high cyclomatic complexity, deep nesting, or excessive dependencies become immediately visible during analysis.

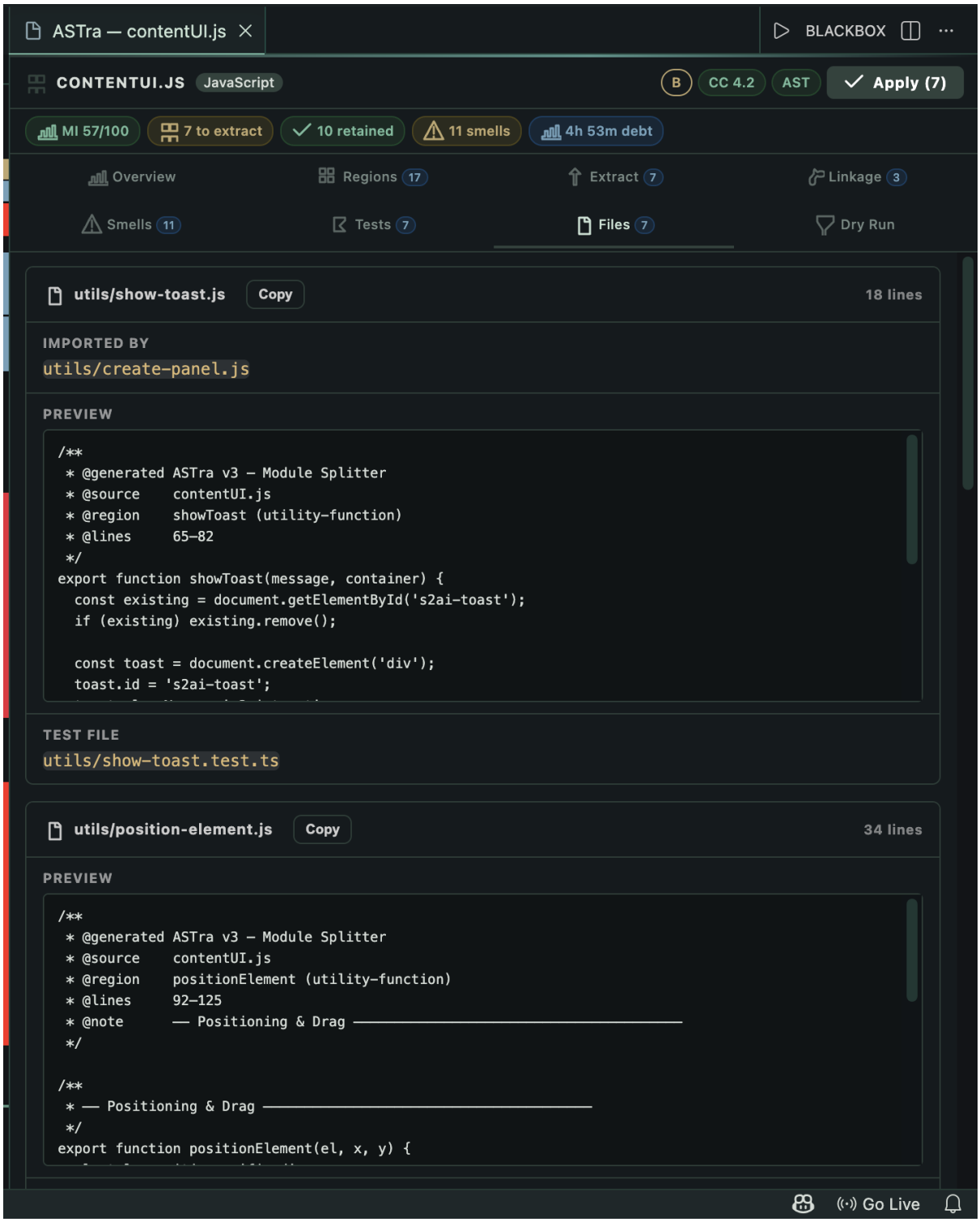
 The smell detection system generates a focused and meaningful set of refactoring suggestions based on detected code quality issues. Instead of producing vague recommendations, the tool highlights specific maintainability problems such as large components, long functions, mixed responsibilities, and excessive nesting.


ASTra — contentUI.js
CONTENTUI.JS
JavaScript
B
CC 4.2
AST
Apply (7)
MI 57/100
7 to extract
10 retained
11 smells
4h 53m debt
Overview
Regions 17
Extract 7
Linkage 3
Smells 11
Tests 7
Files 7
Dry Run
HEALTH GRADE
B
METRICS
LINES OF CODE
451
AVG CC
4.2
AVG COGNITIVE CC
6.2
MAINTAINABILITY
57/100
HALSTEAD VOLUME
1240
TECH DEBT
293 min
BUNDLE WEIGHT
449
DUP LOGIC RISK
100%
HALSTEAD-CALIBRATED THRESHOLD
σ_threshold
10%
FILE PROFILE
Highly Complex File
HALSTEAD P75 EFFORT
30210
USER BIAS
+0%
EXPLANATION
File is highly complex (P75 effort: 30210). Threshold lowered to 10% — aggressive extraction recommended.
INCREMENTAL CACHE
Cache Performance
0% hits

2. Observations on the user interface

1. The eight tab webview organizes information in a clear flow from overview to dry run.
2. The Regions tab helps users understand why a region is classified as a component, hook, or utility.
3. The Extract tab provides confidence labels and suggested file names, which lowers decision effort.
4. The Files tab gives a preview of generated files, which reduces fear of unexpected changes.





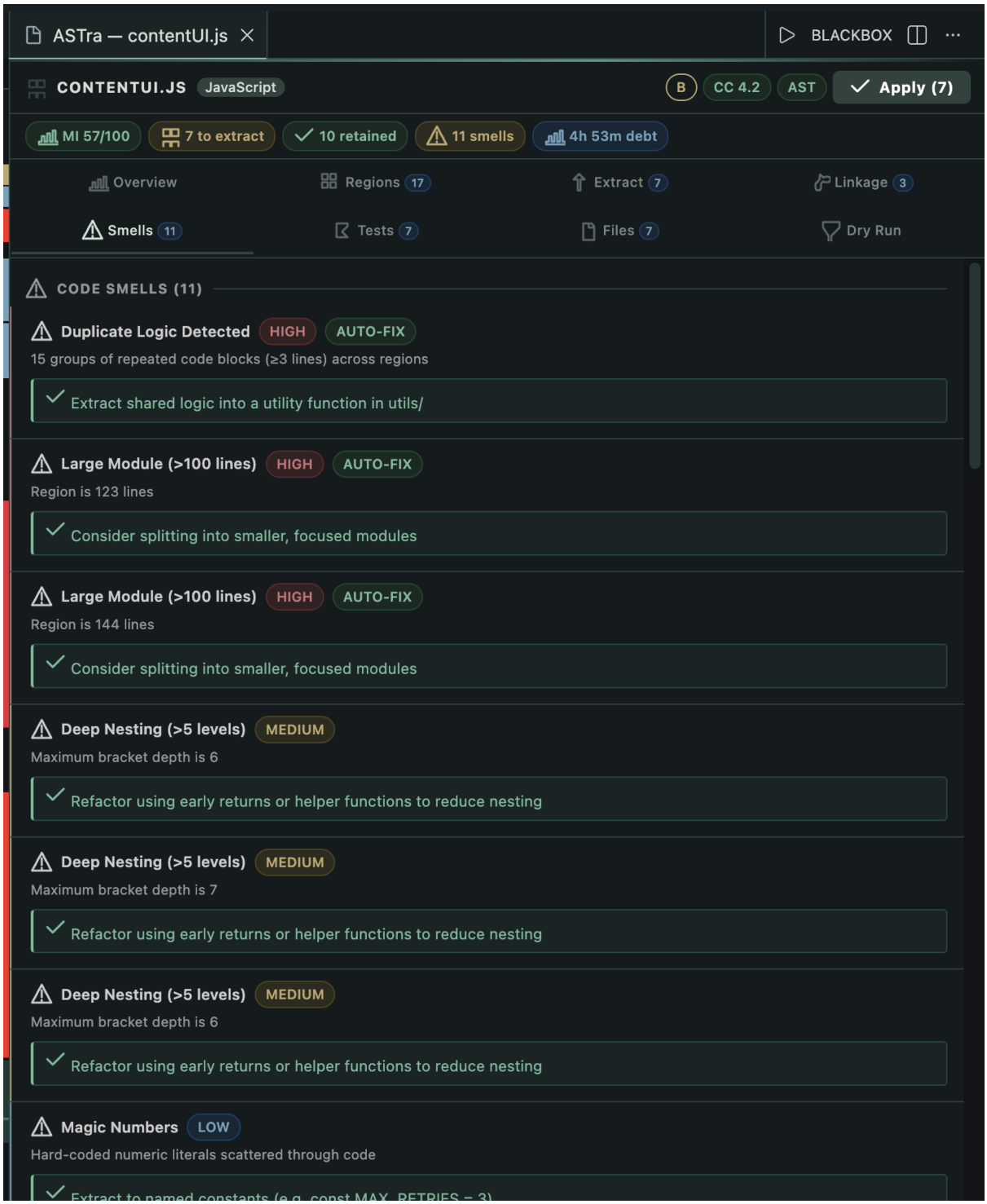
<!-- Screenshot: Files Preview in editor -->

3. Observations in the editor

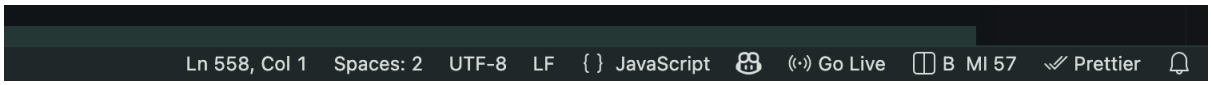
1. Inline diagnostics are useful because they bring smells directly into the editor view.

2. The status bar grade is a quick signal of file health without opening the full panel.

3. The diagnostics use severity levels, which helps distinguish between critical and low level issues.



<!-- Screenshot: Inline smell squiggles in editor -->



<!-- Screenshot: Status bar health grade -->

4. Observations on performance

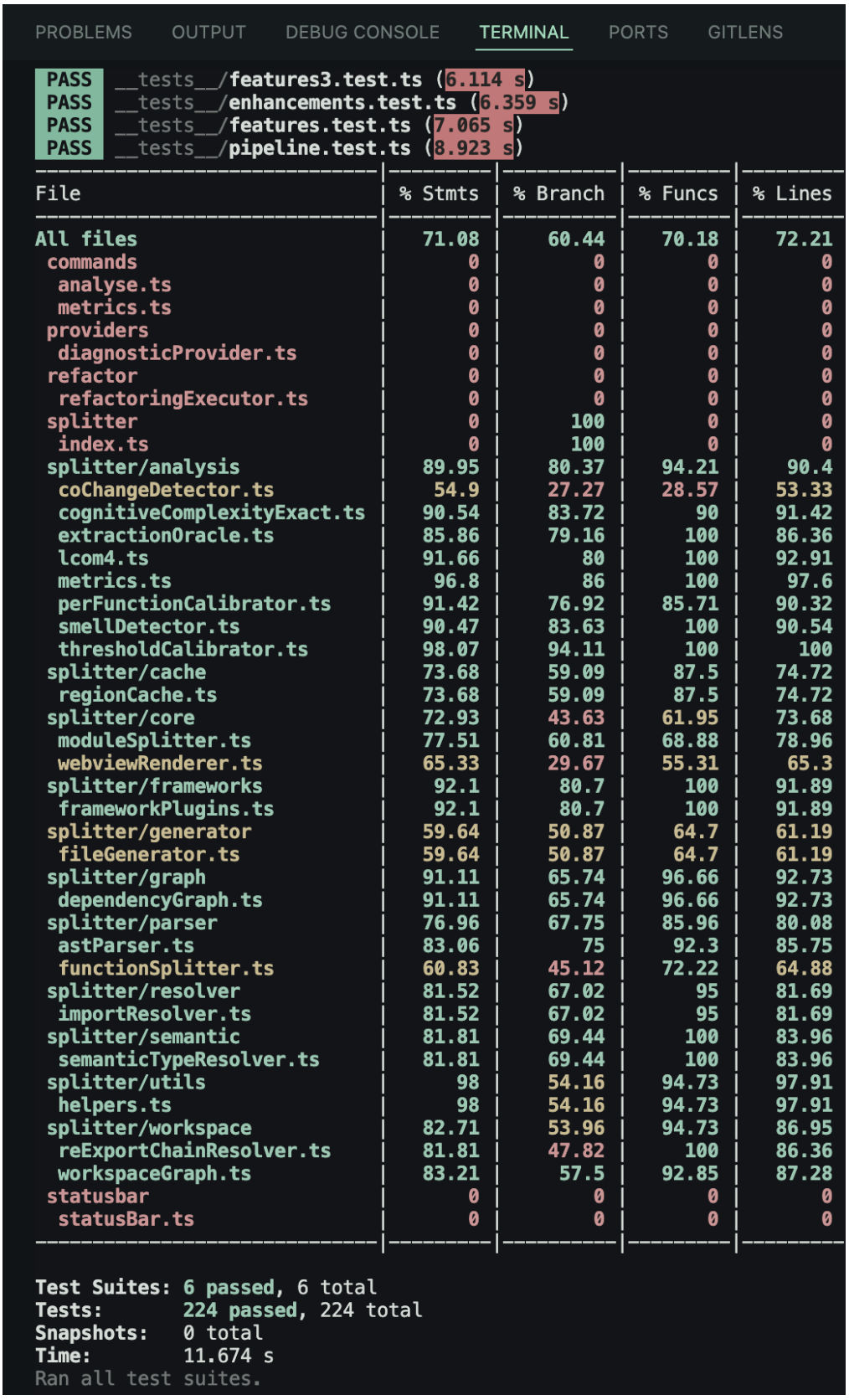
1. The incremental cache reduces repeated analysis time for unchanged regions. This makes the tool usable in real projects where analysis may be run multiple times on the same file.
2. When only a small change is made, only the affected region is re analyzed, and the rest of the analysis uses cached data.
3. The dependency graph is rebuilt when the file hash changes, which ensures correctness even when the file structure changes.

5. Observations on extraction safety

1. The apply step is atomic, so all file creation and updates are done together.
2. The Undo All action is available immediately after apply, which reduces the risk of accidental refactor.
3. The generated file content preserves the original region body, so the behavior of code does not change.

6. Observations on testing

1. The test suite is based on Jest and covers pipeline, metrics, smells, and framework plugins.
2. Test stubs are generated for new files, which encourages writing proper tests after refactoring.



<!-- Screenshot: Jest test run output -->

7. Observations on limitations

- When the analyze command is executed, the extension immediately displays a progress notification to inform the developer that the analysis process has started. This feature is particularly useful because large TypeScript or React files may require additional time for parsing, dependency analysis, and metric computation.
- The tool successfully identifies different logical regions inside the source file such as React components, custom hooks, utility functions, constants, interfaces, helper methods, and type definitions. This observation shows that the AST-based analysis architecture can accurately understand the structure of modern frontend applications.
- The dependency graph subsystem clearly identifies relationships between regions and determines which parts of the code rely on others. This is important because some functions, hooks, or variables are tightly connected and should remain together during extraction.
- The metrics engine provides a quick and structured overview of code complexity, maintainability, coupling, and size-related characteristics. Regions with high cyclomatic complexity, deep nesting, or excessive dependencies become immediately visible during analysis. This helps developers quickly identify problematic sections of the file that may require restructuring or modularization.
- The smell detection system generates a focused and meaningful set of refactoring suggestions based on detected code quality issues. Instead of producing vague recommendations, the tool highlights specific maintainability problems such as large components, long functions, mixed responsibilities, and excessive nesting.

RESULT AND DISCUSSION

This section presents the main results of the project and discusses their meaning, strengths, and limitations.

1. Results

1.1 Functional outcomes

The It Module Splitter delivers the following functional results:

- It detects logical regions such as components, hooks, utilities, classes, and types.
- It computes metrics for each region and for the overall file.
- It detects a wide set of code smells and presents them with severity and recommendations.
- It builds a dependency graph and identifies cycles and extraction order.
- It decides which regions should be extracted using a multi factor score.
- It generates new files, updates the original file, and creates a barrel index.
- It generates test stubs aligned to the region kind.
- It provides a webview panel that presents all results in an understandable way.

1.2 Example style results

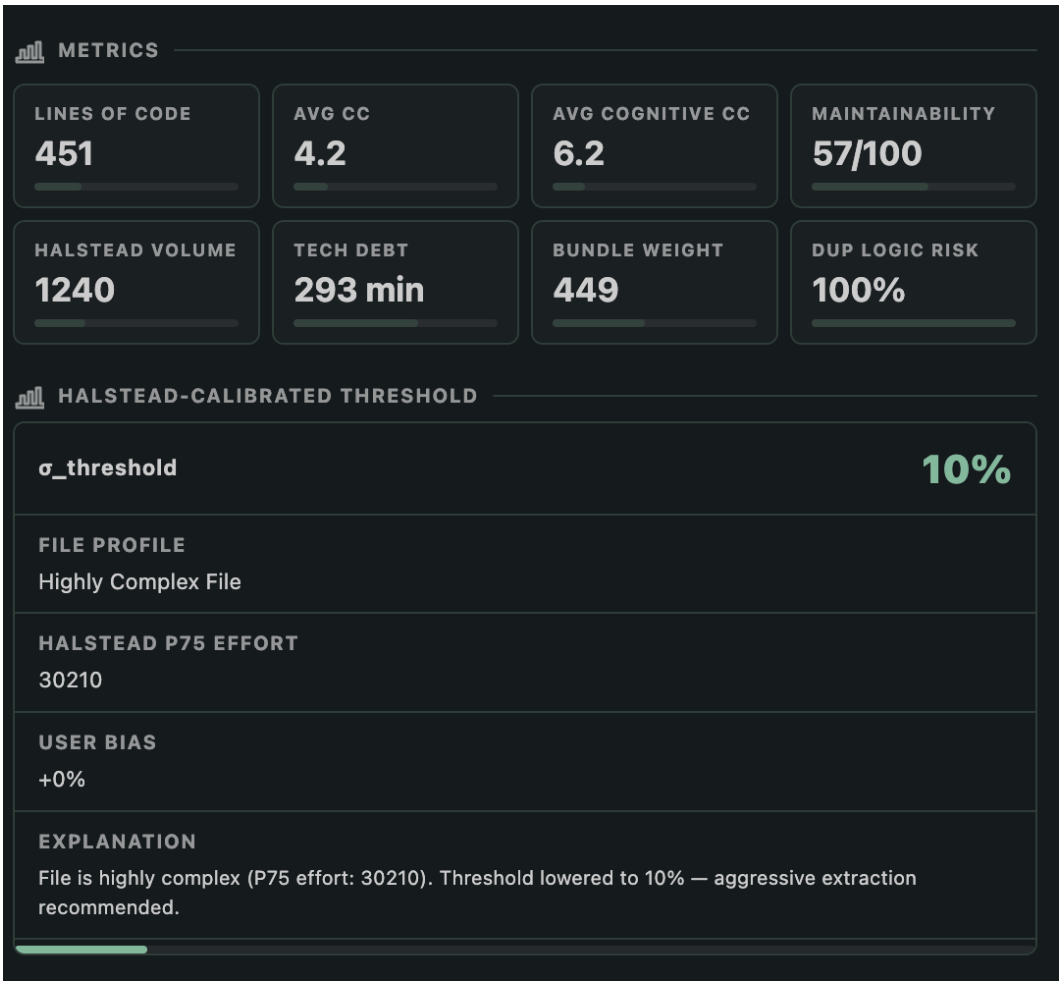
A common example is a large dashboard file that contains a component, a hook, a utility, and some types. The expected outcome after analysis is:

- The component and the hook are recommended for extraction.
- The utility is retained if it is small and pure.
- The types are routed to a types file instead of a new file.
- The updated source file becomes shorter and more focused.

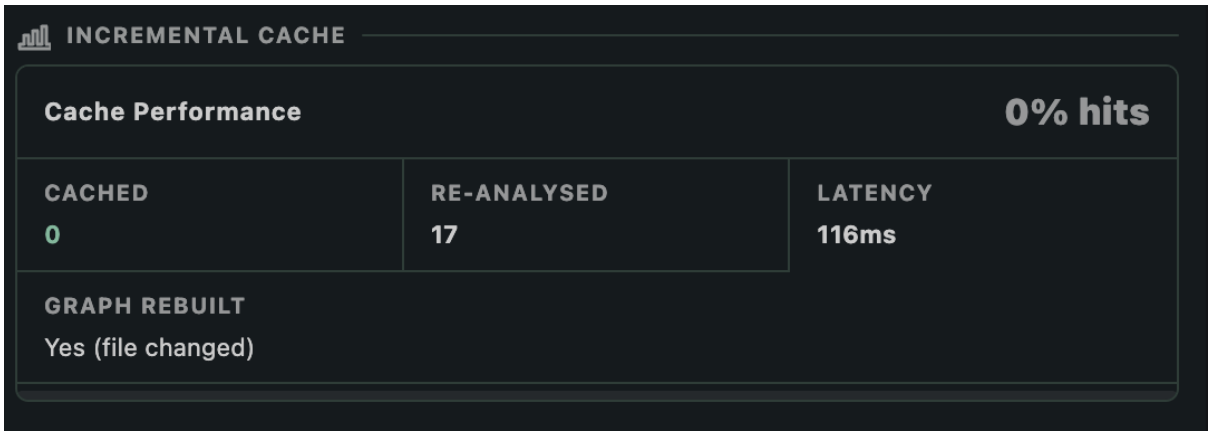
These results match the intended behavior described in the design and documentation.

1.3 Visual output

The webview panel presents the analysis results in eight tabs. This provides a clear visual understanding of the refactor before any change is applied.



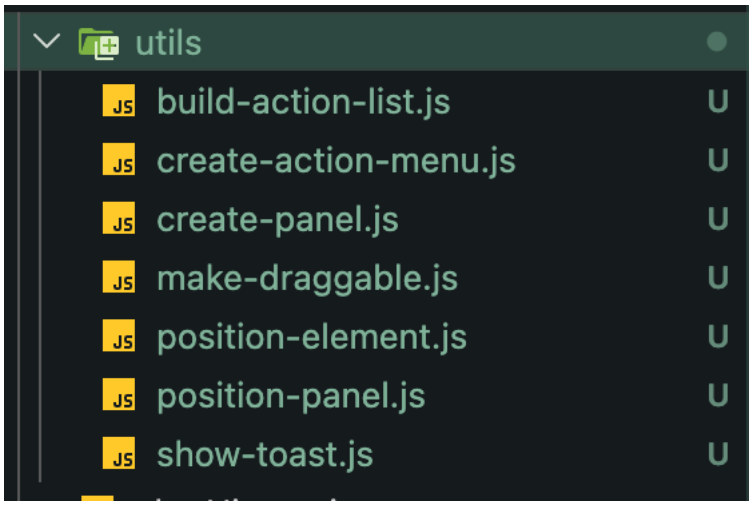
<!-- Screenshot: Webview Overview with metrics and cache card -->



<!-- Screenshot: Webview Linkage tab showing file relationships -->

1.4 Generated files

When a split plan is applied, new files are created with complete imports and preserved code. The tool also creates an index file for easy exports. This is a tangible result that developers can use immediately.



<!-- Screenshot: Generated files in the file explorer -->

1.5 Test outputs

The project includes a Jest based test suite that validates the analysis pipeline. Running the tests produces a clear pass or fail report, which can be included as evidence of correctness.

```
Test Suites: 6 passed, 6 total
Tests:       224 passed, 224 total
Snapshots:   0 total
Time:        11.674 s
Ran all test suites.
```

<!-- Screenshot: Jest test run output showing passing tests -->

2. Discussion

2.1 Impact on maintainability

The primary impact is on maintainability. Large files are split into smaller files that each represent a single responsibility. This improves readability and makes testing easier. The maintainability index and cognitive complexity metrics provide numerical support for this improvement.

2.2 Decision transparency

The Extraction Oracle provides a confidence level and a score for each extraction candidate. This makes the tool trustworthy because the user can understand why a suggestion is made. The system avoids black box decisions and makes refactoring a guided process.

2.3 Safety of changes

The atomic apply using WorkspaceEdit is important for safety. Without this, a partial refactor could leave the project in a broken state. The single undo action gives the user confidence to try the split without fear of permanent damage.

2.4 Performance and usability

The incremental cache reduces repeated work and makes the tool responsive. This is important because developers often iterate on a file and run analysis multiple times. The status bar grade and inline diagnostics provide lightweight feedback without forcing the user to open the full panel every time.

2.5 Limitations

A. Single-File Scope

Each invocation of the analysis pipeline processes one file. The workspace graph provides cross-file symbol visibility for merge suggestions and re-export resolution, but the dependency graph and extraction oracle operate on regions within the current file only. A region that appears simple in isolation but is tightly coupled to many external services may receive a lower coupling score than deserved.

B. Semantic Type Resolution Cost

The `SemanticTypeResolver` invokes `ts.createProgram`, which is expensive on first call (150–400ms for a typical project with dependencies). The 20-entry program cache amortises this cost for repeated analysis, but cold starts on large projects remain noticeable. A future optimization is to reuse the TypeScript Language Server's already-running type-checker instance available via VS Code's extension API.

C. Framework Plugin Coverage

The three framework plugins (Vue, Angular, Svelte) use regex-based block extraction rather than framework-specific parsers. Vue SFC templates are not parsed by a Vue-aware parser; Angular template syntax inside component decorators is treated as a plain string. This means template-level smells have lower precision than script-level smells.

D. Git History Integration

The current implementation does not read version control history. Smells that are best detected through change frequency (e.g. a region that changes in every commit) or co-change patterns (two regions that always change together) are not detected.

CONCLUSION AND FUTURE SCOPE

Conclusion

It Module Splitter solves a practical and common problem: large TypeScript or React files that are hard to maintain. The project uses a well designed pipeline of parsing, graph analysis, metrics, smell detection, and a decision engine to produce a split plan that is both accurate and explainable. It integrates directly into VS Code, which makes it usable for real development and for academic demonstrations.

The system is not only about splitting files. It provides a structured review of code quality through metrics and smells, and it gives the user confidence with a clear webview and an atomic apply. The design keeps the tool safe, and the incremental cache makes it fast enough to use repeatedly. Overall, the project demonstrates that code quality research can be applied to real world developer workflows.

Future scope

The project roadmap already identifies several improvements that can further increase accuracy and usability. The most important future scope items are listed below.

1. Accuracy improvements

- Decorator aware region boundaries to avoid leaving class decorators in the source file when a class is extracted.
- Better classification of chained call constants so that query builders are treated like utilities.
- Support for multiple declarations in a single variable statement to avoid missing small utilities.
- JSX based component detection even when the function name is lowercase.
- Better detection for default export arrow functions and correct naming.

2. Dependency and type resolution

- More precise separation of type only dependencies to avoid extra runtime imports.
- Cross file symbol resolution that follows re export chains.
- A full semantic type checker to improve correctness of imports.

3. Post apply validation

- Run tsc --noEmit automatically after apply to catch any import errors early.
- Optional test run before apply, with an abort if tests fail.

4. Smarter decision making

- Machine learning assisted extraction suggestions, while still keeping an explainable baseline.
- AI assisted file naming based on project conventions.
- Automated cycle breaking recommendations for circular dependencies.

5. Ecosystem expansion

- A Language Server Protocol version to support other editors.
- GitHub Actions integration for CI quality gates.
- Deeper framework plugins that can detect more advanced patterns in Vue and Angular.

6. User experience

- Real time analysis on typing with safe debounce.
- Improved line number precision for framework smells.
- Better visualization of dependency graphs in the webview.

The current system is already functional and useful. The future scope focuses on improving accuracy, automation, and cross editor availability without losing the simple and safe workflow that makes the tool practical.

REFERENCES

[1] T. J. McCabe, "A complexity measure," *IEEE Transactions on Software Engineering*, vol. SE-2, no. 4, pp. 308–320, Dec. 1976.

[2] M. Shepperd, "A critique of cyclomatic complexity as a software metric," *Software Engineering Journal*, vol. 3, no. 2, pp. 30–36, Mar. 1988.

[3] G. K. Gill and C. F. Kemerer, "Cyclomatic complexity density and software maintenance productivity," *IEEE Transactions on Software Engineering*, vol. 17, no. 12, pp. 1284–1288, Dec. 1991.

[4] M. H. Halstead, *Elements of Software Science*. New York, NY, USA: Elsevier, 1977.

[5] C. A. Fenton and S. L. Pfleeger, *Software Metrics: A Rigorous and Practical Approach*, 2nd ed. London, UK: International Thomson Computer Press, 1997.

[6] P. Oman and J. Hagemeister, "Metrics for assessing a software system's maintainability," in *Proc. Conf. Software Maintenance*, Orlando, FL, USA, Nov. 1992, pp. 337–344.

[7] G. A. Campbell, "Cognitive Complexity: A new way of measuring understandability," SonarSource S.A., Geneva, Switzerland, White Paper, 2018. [Online]. Available: https://www.sonarsource.com/docs/CognitiveComplexity.pdf

[8] M. Hitz and B. Montazeri, "Measuring coupling and cohesion in object-oriented systems," in *Proc. Int. Symp. Applied Corporate Computing*, Monterrey, Mexico, Oct. 1995, pp. 25–27.

[9] R. E. Tarjan, "Depth-first search and linear graph algorithms," *SIAM Journal on Computing*, vol. 1, no. 2, pp. 146–160, Jun. 1972.

[10] A. B. Kahn, "Topological sorting of large networks," *Communications of the ACM*, vol. 5, no. 11, pp. 558–562, Nov. 1962.

[11] E. Murphy-Hill and A. P. Black, "An interactive ambient visualization for code smells," in *Proc. 5th Int. Symp. Software Visualization (SOFTVIS)*, New York, NY, USA, 2008, pp. 45–58.

[12] X. Ge, S. Sarkar, and R. Murphy, "Reconciling manual and automatic refactoring," in *Proc. 34th Int. Conf. Software Engineering (ICSE)*, Zurich, Switzerland, Jun. 2012, pp. 211–221.

[13] Facebook Inc., "jscodeshift: A JavaScript codemod toolkit," GitHub Repository, 2019. [Online]. Available: https://github.com/facebook/jscodeshift

[14] D. Sherrets, "ts-morph: TypeScript Compiler API wrapper," GitHub Repository, 2023. [Online]. Available: https://github.com/dsherret/ts-morph

[15] B. Straub, "eslint-plugin-import: ESLint plugin with rules to help with ES2015+ imports," GitHub Repository, 2022. [Online]. Available: https://github.com/import-js/eslint-plugin-import

[16] Nrwl Inc., "Nx: Smart Monorepos · Fast CI," Documentation, 2024. [Online]. Available: https://nx.dev

[17] Vercel Inc., "Turborepo: The build system that makes ship happen," Documentation, 2024. [Online]. Available: https://turbo.build/repo

[18] D. J. Bernstein, "Hash function djb2," cr.yp.to, 1991. [Online]. Available: https://cr.yp.to/cdb/cdbmake.html

[19] A. Tomas, "Managing subscriptions and preventing memory leaks in Angular," *Angular Blog*, Google LLC, Mountain View, CA, USA, Jul. 2019. [Online]. Available: https://blog.angular.io/managing-subscriptions-and-preventing-memory-leaks-in-angular-apps

[20] M. Fowler, *Refactoring: Improving the Design of Existing Code*, 2nd ed. Boston, MA, USA: Addison-Wesley, 2018.

[21] R. C. Martin, *Clean Code: A Handbook of Agile Software Craftsmanship*. Upper Saddle River, NJ, USA: Prentice Hall, 2008.

[22] S. R. Chidamber and C. F. Kemerer, "A metrics suite for object-oriented design," *IEEE Transactions on Software Engineering*, vol. 20, no. 6, pp. 476–493, Jun. 1994.

[23] J. M. Bieman and B.-K. Kang, "Cohesion and reuse in an object-oriented system," *ACM SIGSOFT Software Engineering Notes*, vol. 20, no. SI, pp. 259–262, Aug. 1995.

[24] Microsoft Corporation, "TypeScript Language Specification," GitHub Repository, Version 5.4, Mar. 2024. [Online]. Available: https://github.com/microsoft/TypeScript

[25] Microsoft Corporation, "VS Code Extension API — WorkspaceEdit," Documentation, 2024. [Online]. Available: https://code.visualstudio.com/api/references/vscode-api

[26] T. J. Saaty, "The Analytic Hierarchy Process," *European Journal of Operational Research*, vol. 48, no. 1, pp. 9–26, Sep. 1990.

[27] N. Fenton and J. Bieman, *Software Metrics: A Rigorous and Practical Approach*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2014.

[28] N. Kumar, "module splitter: Adaptive Semantic Tree Restructuring for TypeScript Module Decomposition," GitHub Repository, 2026. [Online]. Available: https://www.github.com/BinaryGuild/module-splitter